

Forcing the Rest

The field equations, the last conditional made unconditional, the Planck-to-cosmological hierarchy in closed form, anomaly freedom by summation, a vacuum angle that has to vanish, and the one place the construction is exposed.

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SECTION ONE

The field equations

Paper 25 identified the action and fixed every relative coefficient. Nobody had varied it, and a Lagrangian nobody varies is an expression rather than a dynamics. Varying it is not hard, because the identification with MacDowell-Mansouri does the work: that variation is known, and the framework inherits it along with the form.

Vary with respect to the coframe and the connection separately. The coframe variation returns the Einstein equations with a cosmological term. The connection variation returns the vanishing of torsion, which is not imposed here but comes out.

$$\mathbf{R}_{ab} - (1/2) \mathbf{R} g_{ab} + \Lambda g_{ab} = \mathbf{T}_{ab} \quad \text{and} \quad \text{torsion} = 0$$

with the cosmological constant already fixed by Paper 24 at three over root five in carrier units, and the stress tensor supplied by the cross term of Paper 25's expansion. Every coefficient in these equations is a ratio of the three weights and none of them is adjustable.

Two consequences are worth stating because they are not choices. Torsion vanishes rather than being set to zero, which is the usual first-order result and which the framework therefore does not have to assume. And the cosmological term is present in the field equations from the start rather than added: it is the held share of the same square whose mixed share is gravity itself. A universe with gravity and no cosmological term would be a square missing a term.

PROVED given the identification of Paper 25, which is where the weight of this section actually rests. The variation itself is standard and is borrowed with its name attached.

SECTION TWO

The last conditional, removed

Paper 21's results carried one condition through four papers: that the reduction averages over a reflection subgroup, which held provided the decayed sublattice was generated by roots. Testing that properly showed rootless rank-three subspaces of E8 are the common case rather than the exception, so the condition looked genuine and was recorded as genuine.

It is not, and the reason was already in the corpus two printings before the question arose. Result thirteen does not say the readable object is a subspace. It says it is the physical algebra. And axiom six's bimodularity makes the reduction's image an algebra rather than merely a subspace. So the surviving block is the Cartan of a reductive subalgebra of rank five, and the question is not whether a rootless subspace exists but whether one can carry a gauge algebra.

rootless rank-three subspaces sampled	:	15893 of 19961
most gauge roots any rootless one leaves	:	20
rootless cases leaving a D5-sized algebra (40)	:	0
the A3 that Paper 21 forced leaves	:	40 (D5 = so(10))

None of them. With no roots in the decayed part there are too few left orthogonal to it to build a rank-five reductive algebra from, and the best any rootless candidate manages is half of what is needed. A rootless decay produces an abelian remnant, which is not a gauge algebra and is not what result thirteen asks for.

the decayed part must be root-generated, and T13 is why

So the condition is discharged, and it is discharged by a result the corpus proved for entirely different reasons before anyone needed it here. Paper 21's four dependent results, the spatial rank, the gauge algebra, the matter modules and the family count, lose the qualification they have been carrying.

PROVED given T13 and A6, with the standard results cited rather than recomputed: a norm-one projection onto a subalgebra is a conditional expectation, and the covariant conditional expectation onto a finite-group fixed algebra is unique and is the average. Certified by scripts/forcing_the_rest.py and scripts/o1_closure.py.

SECTION THREE

The hierarchy, in closed form

The ratio between the cosmological scale and the Planck scale is one of the largest pure numbers in physics and is usually treated as something that has to be explained. The framework reaches the cosmological constant twice, once from the three weights and once from one retention per root over a complete closure, and the two together fix the ratio with nothing left over.

e11 / l_P from the two routes	=	6.848726411e+60
closed form	=	$\sqrt{3} \sqrt{3} / (4 \pi \sqrt{5}) * r^{-120}$
	=	6.848726411e+60 agree

the exponent is minus one hundred and twenty, which is half the root count

There is nothing to explain. The number is r raised to half the number of primitive crossings the carrier has, times a prefactor built from the covolume and the boundary record. Both were fixed elsewhere for other reasons. The vast ratio is not a coincidence requiring a mechanism; it is a retention taken over half the carrier's crossings, and the only thing that had to be true for it to come out this size is that the carrier has two hundred and forty roots.

PROVED as an identity between the framework's own quantities. PROPOSED as the claim that this constitutes an explanation of the observed hierarchy, since the identification of λ with the cosmological constant remains proposed and the whole result inherits that grade.

SECTION FOUR

Anomaly freedom, by summation

A chiral gauge theory whose anomalies do not cancel is inconsistent, and the corpus owed the check. It is normally cited, since the sixteen of $so(10)$ is known to be anomaly free. Citing it here would be the wrong move, because the point is whether the generation this construction actually produced is anomaly free, not whether a generation with the same name is.

So the sum was taken over the sixteen states built in Paper 25 from the three-two split, with the hypercharges those states carry rather than the ones they are supposed to carry.

states in the even exterior algebra	: 16	
sum of Y over the generation	: 0	gravitational-gauge anomaly
sum of Y^3 over the generation	: 0	hypercharge cubed anomaly

Both vanish. The generation the framework built is anomaly free, and it is anomaly free because the hypercharges are what tracelessness on a three-two split forces, not because anybody arranged it.

SECTION FIVE

A vacuum angle that has to be zero

Paper 25 observed that the framework's action has a universal coefficient on the term whose value is the vacuum angle, so no angle can be made small by tuning. One of them has to be zero for a reason. Here is the reason, stated as an argument rather than as a compatibility.

The vacuum of a gauge theory is a sum over sectors labelled by an integer winding, weighted by a phase which is the vacuum angle times that integer. The winding is an integral of the curvature against itself, and it changes sign when the orientation of the base is reversed. The winding is therefore an orientation-defined label: to know which sector you are in, you must first have an orientation to measure it against.

Paper 22 sorted the sectors by whether their directions can be traversed both ways. The colour block is the three freely reversible modes. A block that carries no orientation record cannot carry a label defined by orientation, so the winding is not a readable label there, so the sum over sectors carries no phase to attach.

the strong vacuum angle is not small. It is absent.

And the same reading requires the opposite in the weak sector, which lives on the clock and the closure, the two directions the framework proves are not reversible. There, orientation is exactly what the block carries, the winding is readable, and a phase is available. CP violation in the weak sector is not permitted by this account; it is demanded by it.

The strong vacuum angle is measured below one part in ten billion and has no accepted explanation. An entire hypothetical particle exists to make it small. On this reading nothing needs to make it small, because there is nothing there to be large.

CONDITIONAL on the reversibility sorting of Paper 22, which is where an attack should be aimed, and which is itself CONDITIONAL on the reading of D18 and T15 recorded there. This is the framework's first statement about an open

problem in physics rather than a reconstruction of a settled one, and it should be treated with the suspicion that deserves until someone computes it rather than argues it.

SECTION SIX

Where the construction is exposed

One result of this pass is not a success and is reported at the same volume as the successes, because a paper that graded its own findings by how much it liked them would be worth nothing.

Result twenty-five makes mass the failure of an operation to commute with the decay. Compute that failure across the matter sector and every root in it returns the same value. There is one number, not three.

distinct projections across the 128 spinor-sector roots : [0.75]

the three families are degenerate at this order

They are not degenerate in the world. The electron and the top quark differ by five orders of magnitude, and the framework as it currently stands says nothing about why. The hierarchy has to come from the Yukawa structure, which this construction does not reach, and which is now the largest single unattacked problem in the corpus.

This is stated plainly because the alternative is worse. A framework that produced three families, stopped, and let the reader assume the masses would follow, would be trading on an expectation it had not earned. Three families is a real result. Three degenerate families is a wrong prediction until the hierarchy is supplied, and the corpus should carry it as an exposure rather than as a pending success.

SECTION SEVEN

The ledger, closed as far as it goes

Discharged this pass The reduction's form, no longer conditional. The field equations, by variation. Anomaly freedom, by summation over the states actually built. The Planck-to-cosmological hierarchy, in closed form as r to the minus half the root count.

Argued this pass The vanishing of the strong vacuum angle, from the reversibility of the colour block. Conditional on Paper 22's sorting and flagged as the most attackable thing in the corpus.

Exposed this pass Family degeneracy. The construction currently predicts three families with the same mass, which is wrong, and the Yukawa structure is the gap.

Still owed The coupling, one debt in three costumes: the fine structure constant, the Rydberg's scale, the normalisation of the Einstein term. The Yukawa structure and the mixing matrices. The uniqueness of the complete carrier closure, finite and not required by anything. And the two measurements, the pipeline audit and the running self-holding system, neither of which closes at a desk.

Four items discharged, one argued, one exposed, and four owed. The four owed are the same four the corpus would have listed a year ago, minus everything that has since been forced, and every one of them now has a stated shape rather than being a direction to look in.

The framework began with a sentence asked to describe itself. It now carries a variational principle whose every relative coefficient is a ratio of two numbers fixed by that sentence, a gauge algebra, a generation with every charge in the right place, three families, a signature, a spectrum, and a hierarchy in closed

form. It does not carry a single mass. That asymmetry is the honest description of where it stands, and the next person to work on it should start there rather than anywhere more comfortable.